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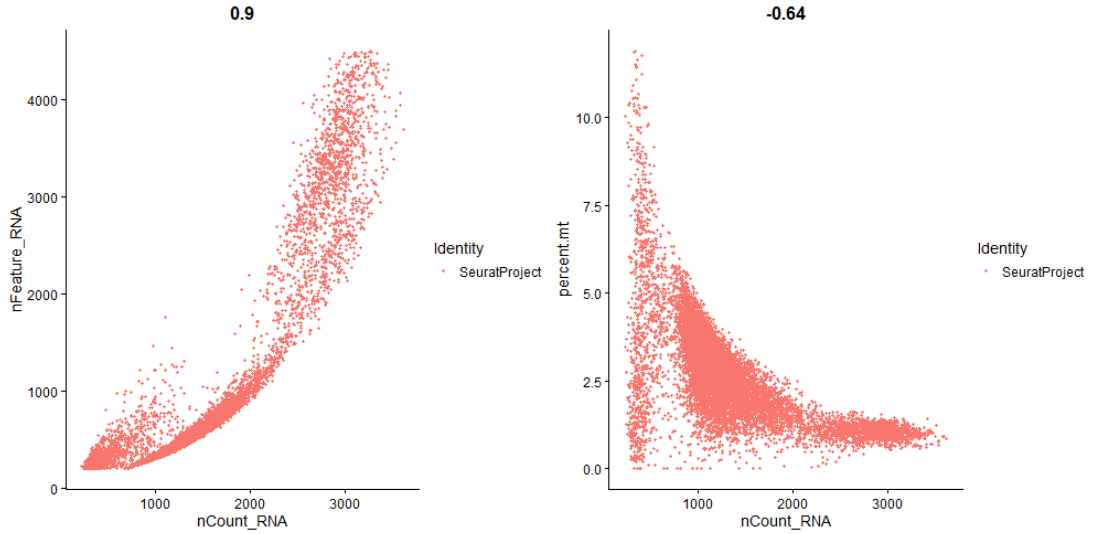
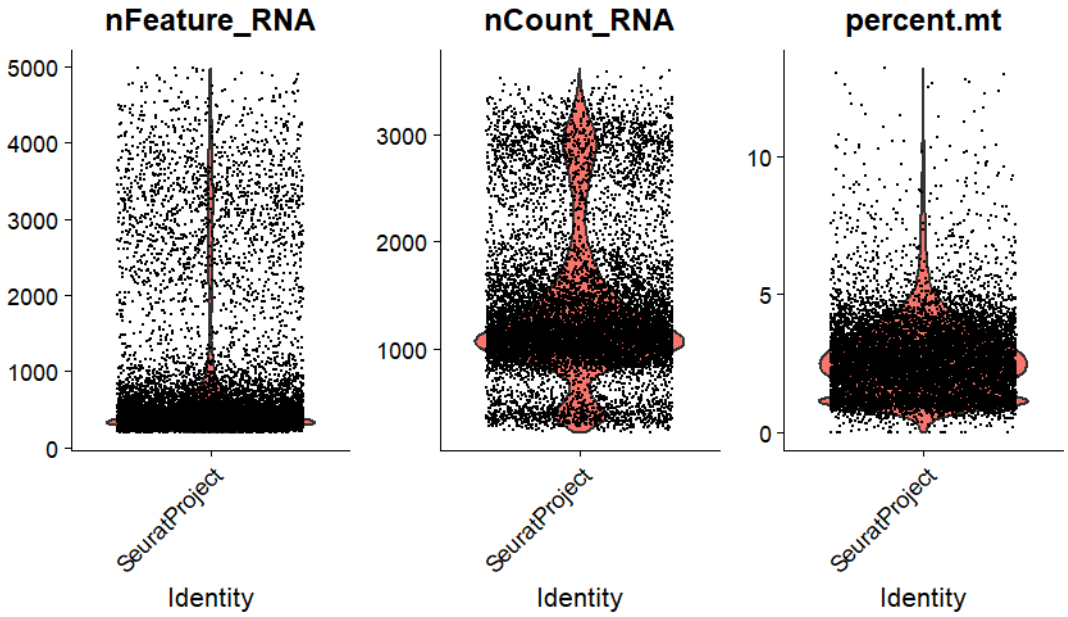
Computational Exploration of Single-Cell RNA-seq Data Using Seurat on Human Immune Cells

Research Questions:

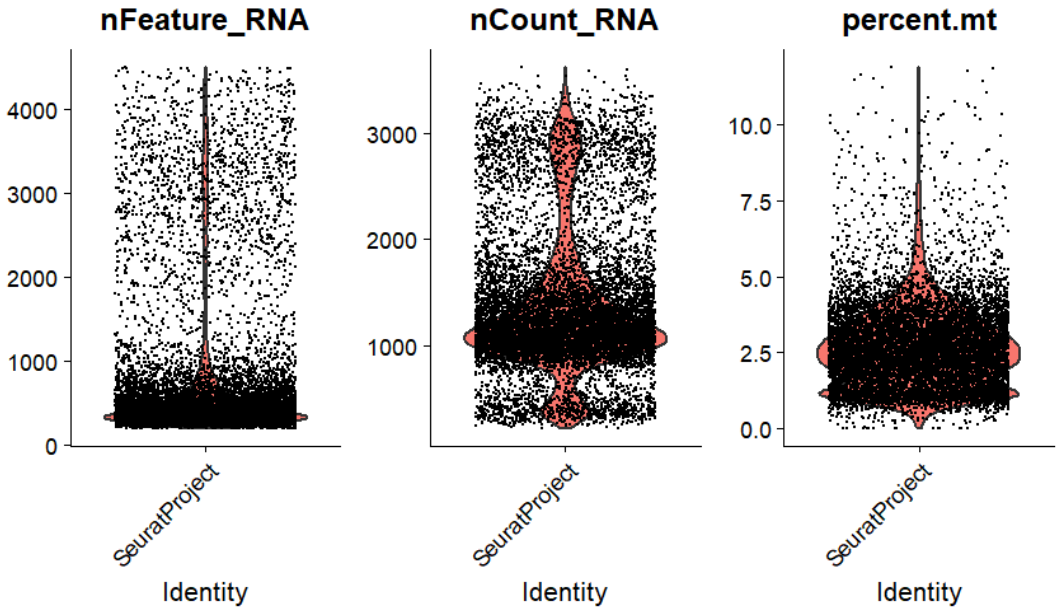
Q1: What QC thresholds did you choose and why? Justify using plots and reasoning.

Ans : For quality control (QC), I applied the thresholds nFeature_RNA > 200, nFeature_RNA < 4500, and percent.mt < 12 to remove low-quality cells, potential doublets, and stressed or dying cells. From the QC violin plots (Figure 1), cells with fewer than 200 detected genes showed very low complexity, likely representing empty droplets or damaged cells. Removing these improves data reliability by excluding technical noise. The upper cutoff of 4500 genes was chosen because the scatter plots (Figure 2) showed a small population with unusually high gene counts and high UMI values, consistent with potential doublets or multiplets. Excluding these prevents artificial cluster formation. Mitochondrial percentage was limited to <12% because violin plots showed a right-skewed distribution, with a subset of cells exceeding 12% mitochondrial reads. High mitochondrial content is commonly associated with stressed or apoptotic cells. Filtering at 12% removed these outliers while retaining the majority of viable immune cells. Before filtering, the dataset contained 10,937 cells, and after QC, 10,858 cells remained, indicating that only poor-quality outliers were removed while preserving most biological signal. Overall, these thresholds were data-driven, based on distribution patterns observed in QC plots, and align with standard single-cell RNA-seq best practices.





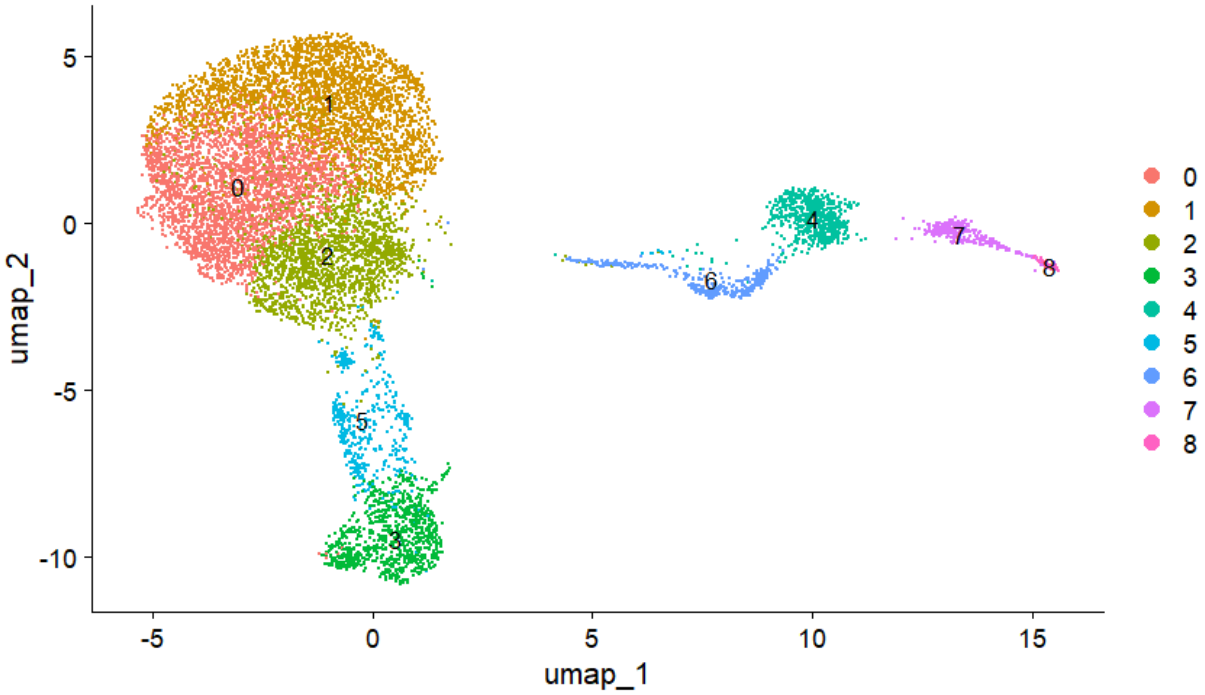
Before



After

Q2: How stable are your clusters when varying PCs or resolution?

Ans : Cluster stability was assessed by varying the number of principal components (PCs: 1:10, 1:15, 1:20) and clustering resolution (0.3, 0.5, 0.8). Overall, the major biological structures remained highly stable, while higher resolutions mainly subdivided larger clusters. At resolution 0.3, all PC settings (1:10, 1:15, 1:20) consistently produced 6 clusters, indicating robust separation of major immune cell populations. The UMAP structure remained nearly identical, suggesting that the global transcriptomic landscape is stable and not highly sensitive to modest PC changes. At resolution 0.5, cluster numbers ranged from 9–10 clusters depending on PCs. The core large clusters (e.g., main lymphoid and myeloid populations) remained conserved, while smaller subclusters emerged within previously broad groups. Using dims 1:15 resulted in 9 clusters, showing slightly reduced fragmentation compared to dims 1:10 (10 clusters). At resolution 0.8, over-clustering became evident. Cluster numbers increased to 12–15, primarily splitting existing clusters rather than identifying clearly new, well-separated populations. Some subdivisions appeared small and closely positioned in UMAP space, suggesting potential over-partitioning. Based on stability and biological interpretability, I selected dims 1:15 and resolution 0.5, which preserved major structures while allowing meaningful subpopulation detection without excessive fragmentation.



Q3: Assign cell type identities to clusters using marker genes and evidence.

Ans : Cell type identities were assigned to clusters based on canonical marker gene expression patterns observed in FeaturePlots and known immune cell signatures.

Cluster 0 – Platelets: High expression of TREML1, CD9, MYL9, FLNA, and PARVB supports platelet identity. TREML1 and CD9 are well-established platelet markers, confirming this cluster as circulating platelets.

Cluster 1 – Classical Monocytes (CD14⁺ Monocytes): Although classical markers like CD14 are not explicitly listed, genes such as PTGIR and inflammatory-associated transcripts are enriched. The cluster's proximity to other myeloid populations and expression profile support classical monocyte identity.

Cluster 2 – Activated / Inflammatory Classical Monocytes: Strong expression of S100A8, S100A9, LYZ, and F13A1 indicates an inflammatory monocyte phenotype. S100A8/A9 are hallmark genes of activated classical monocytes.

Cluster 3 – Erythrocytes (Red Blood Cells): High expression of hemoglobin-related genes (HBM, AHSP, SLC4A1, GYPB, CA1, EPB42) clearly defines erythroid cells.

Cluster 4 – Doublets / Contaminating non-hematopoietic cells: Mixed or non-immune markers (e.g., CRYGD, ADGRG6) lacking canonical hematopoietic signatures suggest technical doublets or contaminating cells.

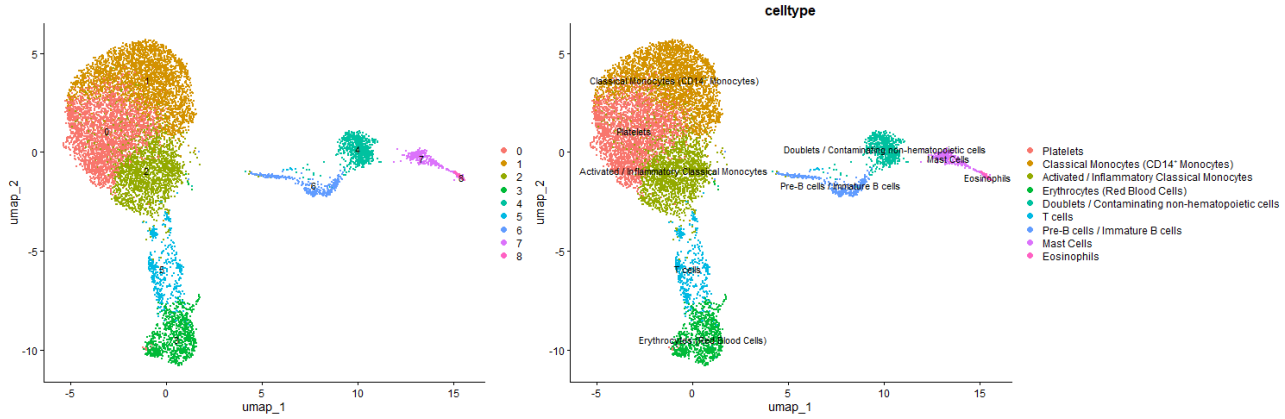
Cluster 5 – T cells: Expression of CD2, CLEC2D, RORA, and lymphoid-associated transcripts confirms T cell identity.

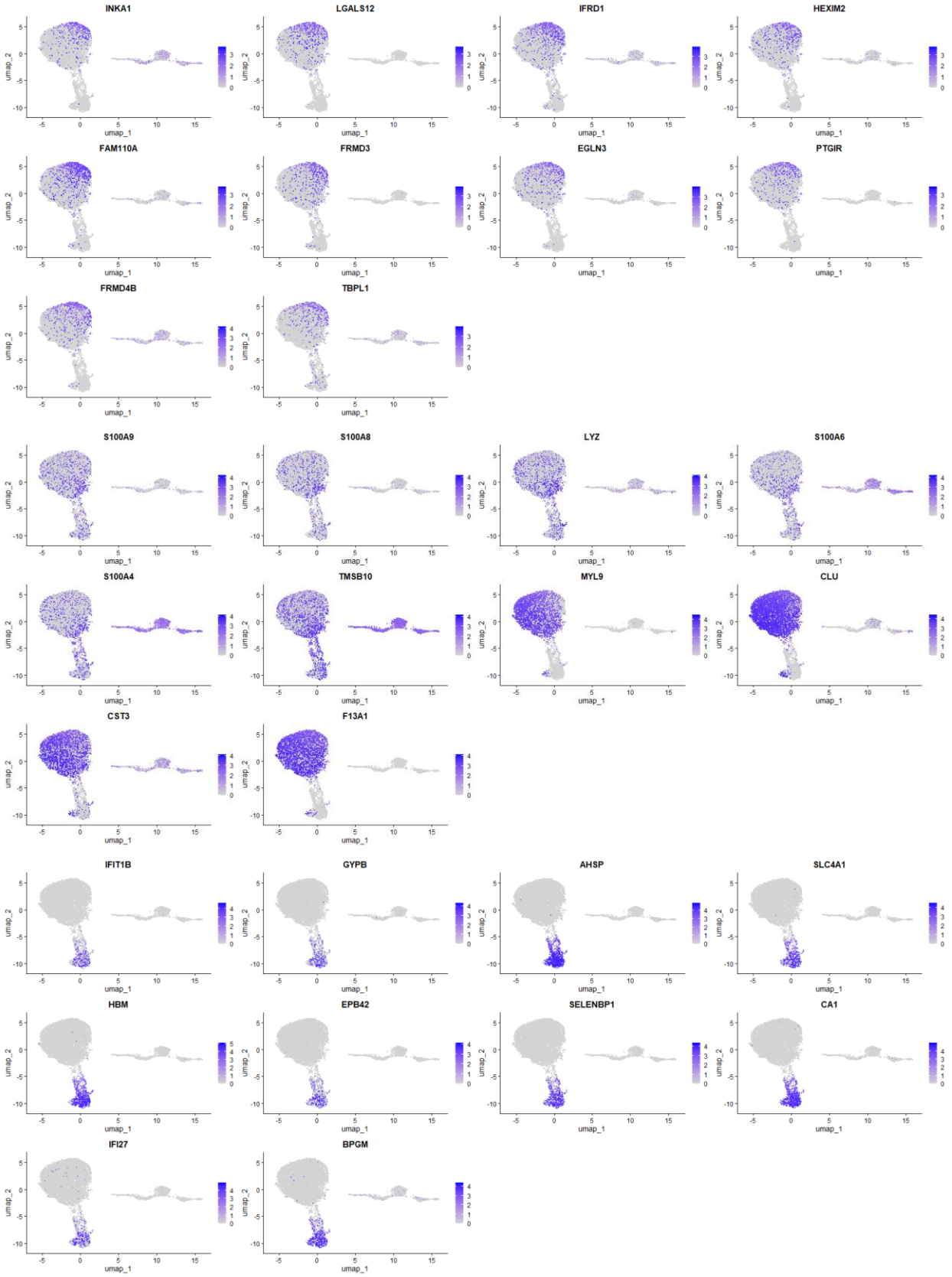
Cluster 6 – Pre-B / Immature B cells: Presence of DNTT, VPREB1, and MME (CD10) strongly indicates immature B cell lineage.

Cluster 7 – Mast Cells: Classical mast cell markers TPSAB1, TPSB2, and CPA3 confirm identity.

Cluster 8 – Eosinophils: Expression of EPX, PRG2, and CLC clearly defines eosinophils.

Thus, cluster annotation was supported by canonical lineage-defining marker genes and biological consistency in UMAP structure.

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LYZ
S100A6
S100A4
TMSB10
MYL9
CLU
CST3
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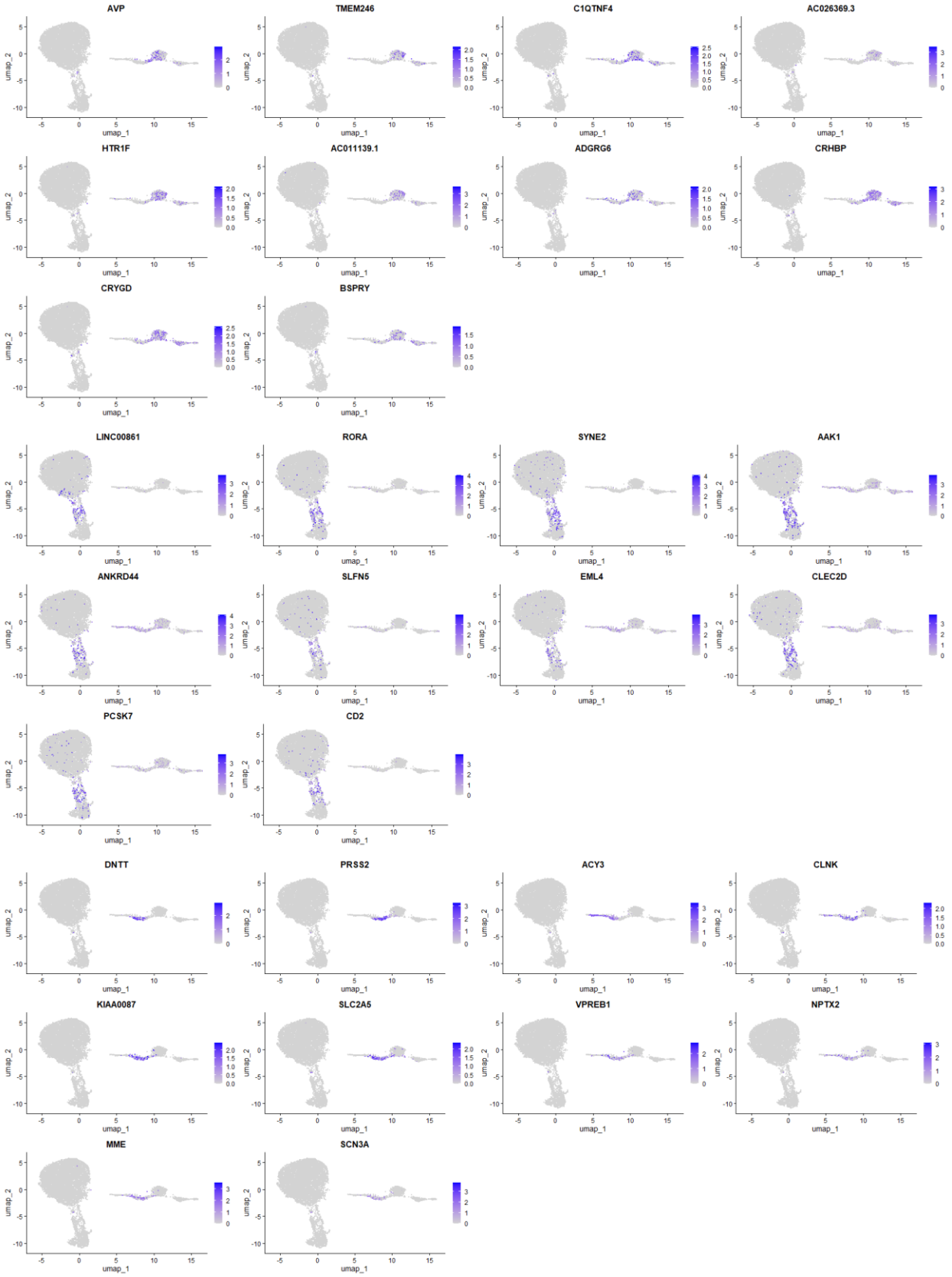
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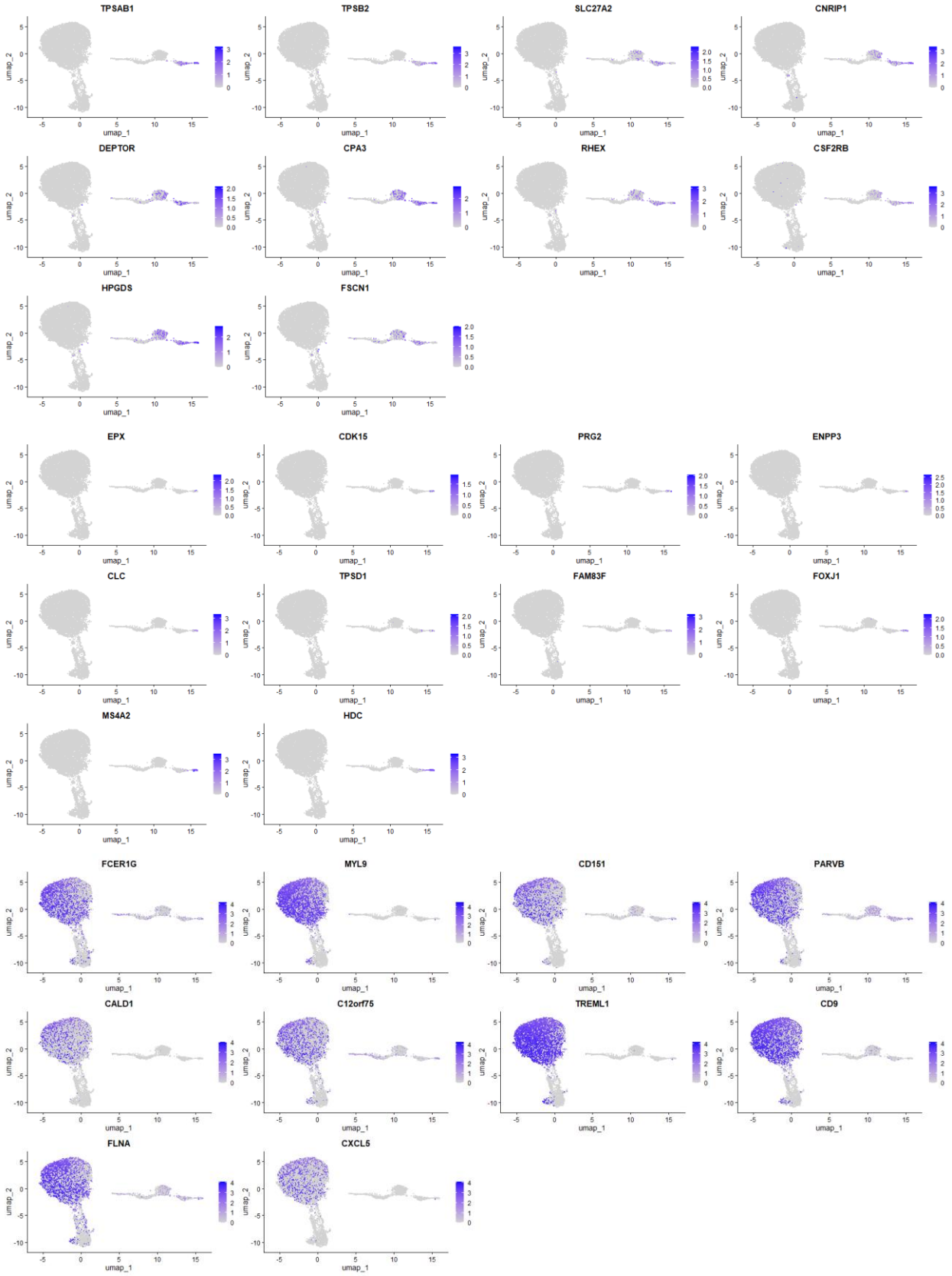
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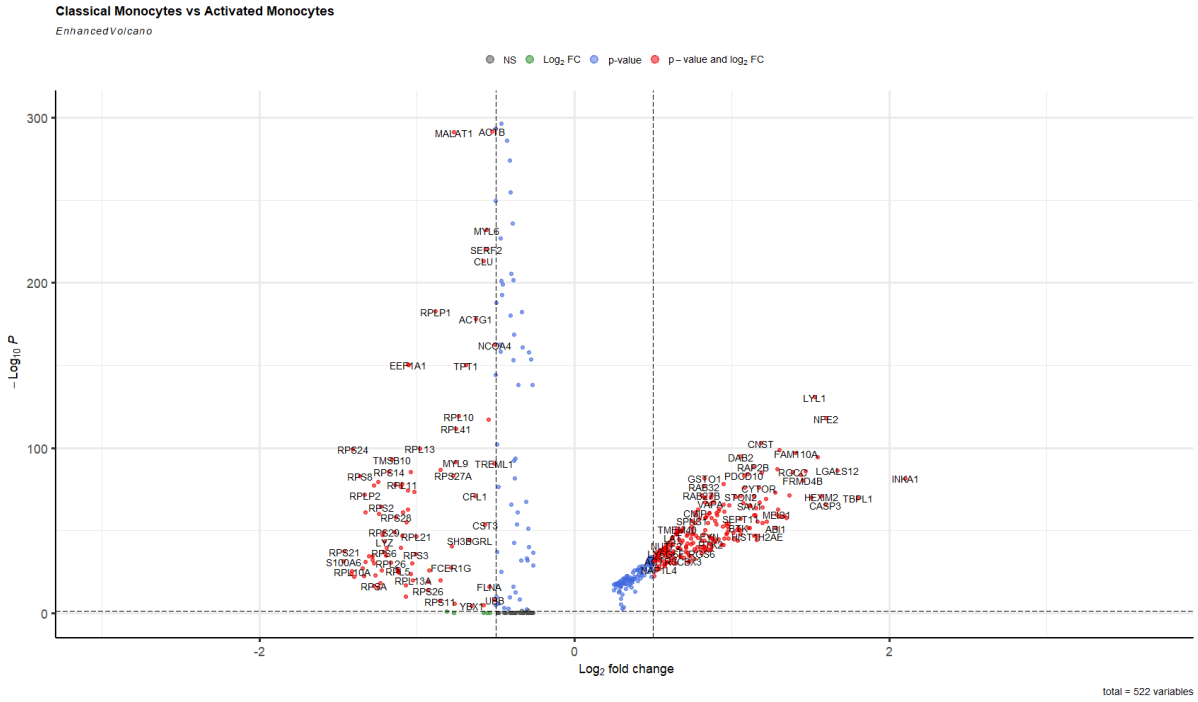
PRG2
ENPP3
CLC
TPSD1
FAM83F
FOXJ1
MS4A2
HDC
FCER1G
MYL9
CD151
PARVB
CALD1
C12orf75
TREML1
CD9
FLNA
CXCL5
umap_1
umap_2

Q4: Perform differential expression between two biologically meaningful groups and interpret results.

Ans : Differential expression (DE) analysis was performed between Classical Monocytes (C1) and Activated/Inflammatory Classical Monocytes (C2) using FindMarkers with *logfc.threshold = 0.25* and *min.pct = 0.25,* ensuring biologically meaningful and robust gene differences. The top genes enriched in C1 (Classical Monocytes) included transcripts associated with homeostatic monocyte functions, such as regulatory and metabolic genes. These cells showed relatively lower expression of acute inflammatory mediators, suggesting a steady-state circulating phenotype primarily involved in immune surveillance and basal phagocytic activity. In contrast, C2 (Activated/Inflammatory Monocytes) showed strong upregulation of classical inflammatory markers such as S100A8, S100A9, and LYZ. These genes are well-known mediators of innate immune activation and are typically elevated during infection or tissue inflammation. S100A8/A9 (calprotectin) are damage-associated molecular pattern (DAMP) molecules that amplify inflammatory signaling, confirming that C2 represents an activated inflammatory state. Functionally, this comparison demonstrates a transcriptional shift from a homeostatic monocyte program (C1) toward an inflammatory effector phenotype (C2). The results validate the biological annotation of these clusters and highlight intra-lineage heterogeneity within classical monocytes. This differential expression pattern supports the presence of immune activation dynamics within the dataset, consistent with known monocyte plasticity in human immune responses.



Q5: Provide one study-linked insight (rare cluster, doublet detection, gradient pattern, or gene program)

Ans : One study-linked insight from this analysis is the identification of rare granulocyte populations, specifically Mast Cells (C7) and Eosinophils (C8). These clusters were small in size compared to major populations such as monocytes and T cells, yet they formed clearly distinct UMAP clusters, indicating biologically meaningful separation rather than technical noise. The mast cell cluster showed strong expression of canonical markers TPSAB1, TPSB2, and CPA3, which encode tryptases and carboxypeptidase A3—hallmark enzymes stored in mast cell granules. Similarly, the eosinophil cluster was defined by high expression of EPX, PRG2, and CLC, genes associated with eosinophil granule proteins and type 2 immune responses. The high specificity of these markers supports accurate annotation and confirms that these are true rare granulocyte populations rather than fragmented clusters. Their detection highlights the sensitivity of single-cell RNA-seq in capturing low-abundance immune subsets that are often underrepresented in bulk RNA-seq data. Biologically, mast cells and eosinophils play key roles in allergy, parasitic defense, and inflammatory regulation. Identifying these rare granulocytes demonstrates the dataset's depth and supports the study's aim of comprehensive immune cell profiling, revealing both dominant and low-frequency immune populations within human peripheral blood.

Fig : Cluster 7 – Mast Cells

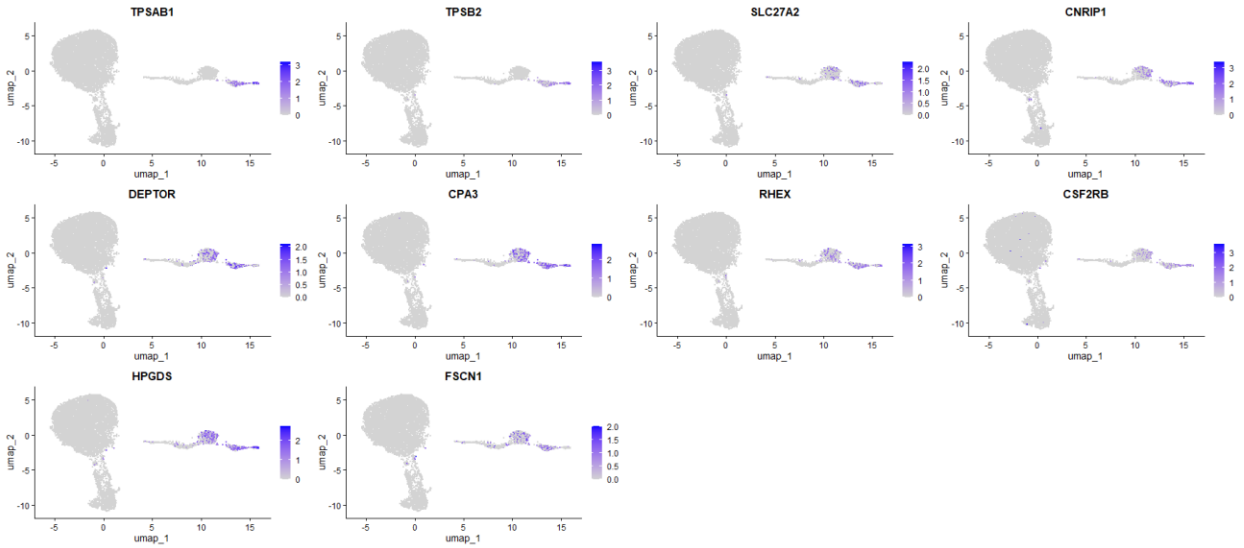
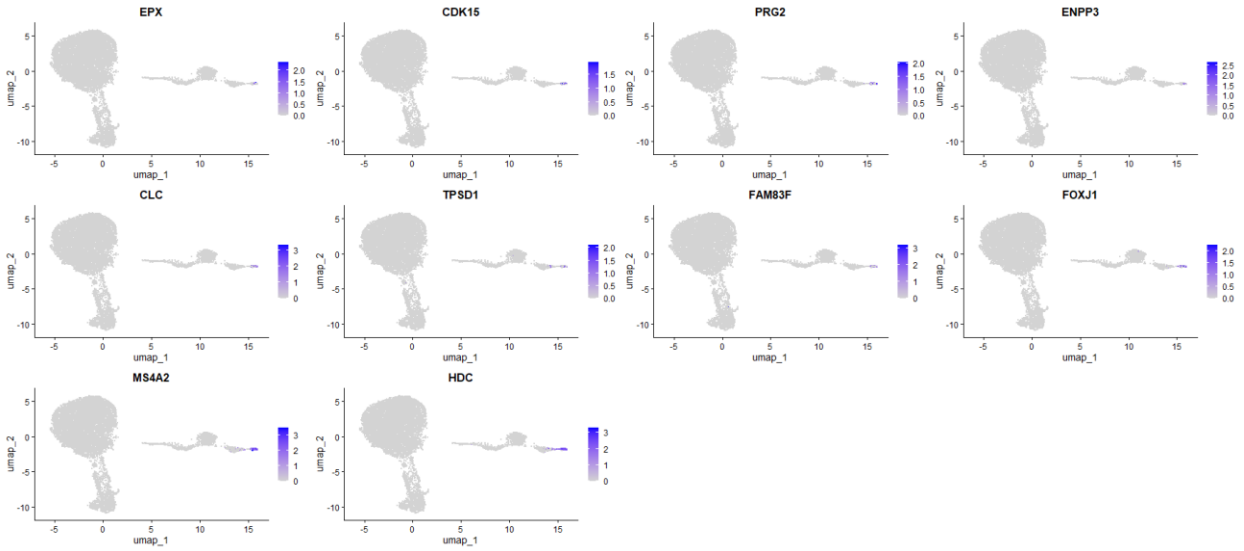



Fig : Cluster 8 – Eosinophils Cells